

Probabilistic Models of the Visual Cortex

Fall 2026 Homework 1

Prof. Alan Yuille

Instructions

1. **Deadline:** Monday, Oct 5, 2026 at 11:59 PM (Eastern Time).
2. **Late homework:** Late homework will not be accepted unless permission is obtained well in advance for documented extenuating circumstances. Email Prof. Yuille for approval and copy the TAs.
3. **Submission:** one PDF file on Gradescope with your written answers and the plots and answers for Questions 8–10, named `<your_jhed>-hw1.pdf`. Code is not required and will not be graded; a notebook export that shows code is fine (File → Save and Export Notebook As → HTML, then print it to PDF from your browser).
4. **Coverage:** Lectures 2–8 (Question 1 also uses Lecture 1, the introduction). 110 points: written Questions 1–7 (72), experimental Questions 8–10 (38).
5. **Warning:** Start the experimental section (Questions 8–10, three notebooks) early! Do not use AI tools (such as ChatGPT, Claude, Gemini or Copilot) to complete this homework. If we detect that AI was used to complete it, you will receive zero points for this homework.
6. **Calculations:** You may use a calculator or Python. Write down the formula and the numbers you plug in, not only the final answer.
7. **Collaboration:** Feel free to discuss the questions with friends, but be sure to write your answers entirely independently!
8. **Questions?** Attend office hours (see the course website) or send an email!

Question 1: Bayes and vision (8 points)

Please give brief and clear answers to the questions below. They are based on the introduction (Lecture 1), the Yuille and Kersten reading (Lecture 2) and Lectures 3 and 4.

1. Vision is extremely hard because natural images are complex and ambiguous. Name *two* sources of complexity (**2 points**) and *two* kinds of ambiguity (**2 points**). Give a concrete visual example for each ambiguity.
2. Human perception can be tested by comparing it with an *Ideal Observer*.
 - (a) What is an Ideal Observer, and how is it used to test human perception? (**1 point**)
 - (b) Yuille and Kersten give two reasons why Ideal Observers usually perform much better than humans. State them. (**1 point**)
3. The Bayes decision rule depends on the image I only through the posterior $P(W | I)$, so a network trained to output $P(W | I)$ directly has everything the decision needs. Give *two* arguments from the lectures for nevertheless modelling the likelihood $P(I | W)$ and the prior $P(W)$. (**2 points**)

Question 2: Perceptual phenomena as Bayesian inference (12 points)

Visual illusions and demonstrations teach us about the assumptions that the brain makes when interpreting images. For each phenomenon below, write a two- to three-sentence explanation of what is perceived and why (**1 point each**). Then name the Bayesian ingredient it illustrates best (**1 point each**), choosing from: *prior*, *generic viewpoint*, *explaining away*, *model selection*, *causal inference*, *strong coupling of cues*, *spatial smoothness prior*. An ingredient may be used more than once. Several phenomena illustrate more than one ingredient: name the one you think fits best and say why in your explanation. All six phenomena are discussed in the Yuille and Kersten reading or in the notes for Lectures 5 and 7.

1. The flying carpet (levitation illusions)
2. The ball in a box (moving cast shadows)
3. The Kanizsa triangle
4. A highlight placed at different depths on a binocularly viewed sphere
5. The ventriloquism effect
6. Random-dot stereograms

Question 3: Bayes decision theory (14 points)

1. (4 points) Concepts.

- (a) What are the formulas for the risk $R(\alpha)$ of a decision rule α and for the Bayes decision rule $\hat{\alpha}$? What is the Bayes risk? **(1 point)**
- (b) What are maximum likelihood (ML) estimation and maximum a posteriori (MAP) estimation? **(1 point)**
- (c) When does the Bayes decision rule reduce to the MAP estimate, and when does it reduce to the ML estimate? **(1 point)**
- (d) What are false positives and false negatives? **(1 point)**

2. (4 points) An edge detector reports its response at three levels, x_1 (weak), x_2 (medium) and x_3 (strong). The joint probabilities of the response and the truth y ($y = +1$: edge, $y = -1$: non-edge) are

	x_1	x_2	x_3
$y = +1$	0.02	0.06	0.12
$y = -1$	0.50	0.24	0.06

A false positive costs 1, a false negative (a missed edge) costs 3, and correct decisions cost nothing. A decision rule α assigns a label to each of the three levels.

- (a) Compute the risk of the rule $\alpha = (-1, +1, +1)$. **(1 point)**
 - (b) Find the Bayes rule and the Bayes risk. Why is the Bayes risk not zero? **(2 points)**
 - (c) For which values of the false-negative cost does the Bayes decision at x_2 become $+1$? **(1 point)**
3. (4 points) The response x of a detector is Gaussian, $\mathcal{N}(2, 1)$ when a target is present ($y = +1$) and $\mathcal{N}(0, 1)$ when it is absent ($y = -1$).
- (a) Show that the log-likelihood ratio is

$$\Lambda(x) = \log \frac{P(x \mid y = +1)}{P(x \mid y = -1)} = 2(x - 1). \quad \textbf{(1 point)}$$

- (b) With equal priors and the 0–1 loss, find the threshold x^* on the response, the hit rate H , the false positive rate F and the Bayes risk. **(1 point)**
 - (c) Now targets are rare, $P(y = +1) = 0.1$, and a miss costs 5 times as much as a false positive. Compute the threshold T on $\Lambda(x)$, the threshold x^* on the response, and the new H and F . **(1 point)**
 - (d) A human observer in a detection task of this kind has $H = 0.75$ and $F = 0.20$. Estimate the observer's sensitivity d' and compare it with the d' of the Ideal Observer. **(1 point)**
4. **(2 points)** The posterior of the depth z of a surface (in metres) is bimodal: $P(z | I) = 0.6\mathcal{N}(z; 2, 0.2^2) + 0.4\mathcal{N}(z; 5, 0.2^2)$.
- (a) Find the MAP estimate, the posterior mean (the Bayes estimate for the squared loss) and the posterior median (the Bayes estimate for the absolute loss). **(1 point)**
 - (b) A robot must reach for the surface. Which estimate would you use, and why is the posterior mean a poor choice here? **(1 point)**

Question 4: Filters and Bayesian edge detection

(6 points)

1. **(2 points)** What kind of image structure does the first-order derivative-of-Gaussian filter respond to, and what does the Laplacian of Gaussian respond to? **(1 point)** Why is the image smoothed with a Gaussian before it is differentiated, and what does the scale σ control? **(1 point)**
2. **(2 points)** Suppose you first smooth an image I with a Gaussian G_{σ_1} and then apply the derivative-of-Gaussian filter $\partial G_{\sigma_2}/\partial x$. Does the order of the two filters matter? Show that the result is a single derivative-of-Gaussian filter applied to I , and give its scale. You may use three facts: convolution is associative and commutative, differentiation commutes with convolution, and convolving two Gaussians gives a Gaussian whose variances add,

$$\frac{\partial}{\partial x}(G * I) = \frac{\partial G}{\partial x} * I, \quad G_{\sigma_1} * G_{\sigma_2} = G_{\sqrt{\sigma_1^2 + \sigma_2^2}}.$$

3. **(2 points)** In an annotated dataset 5% of the pixels are edges, and missing an edge costs 20 times as much as a false positive. Compute the threshold T on the log-likelihood ratio $\log P(f \mid y = +1)/P(f \mid y = -1)$ of the Bayesian edge detector. **(1 point)** What is T if the two kinds of error cost the same, and how does the edge map change? **(1 point)**

Question 5: Probabilities on graphs (14 points)

1. (4 points) **Graphs from Lecture 5.** The three figures below are taken from the notes for Lecture 5.

- (a) Figure 1 shows two ways in which two cues C_1 and C_2 can depend on a scene property S . Which graph shows weak coupling and which shows strong coupling, and what is the difference between them? Under which three conditions is the optimal estimate of S a weighted average of the estimates from the two cues? Give one example of strong coupling from the lectures. (2 points)

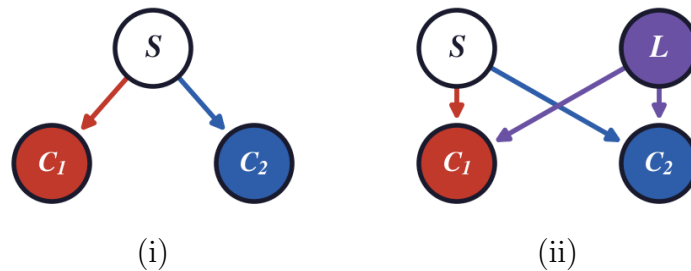


Figure 1: Two graphs relating a scene property S to two cues C_1 and C_2 .

- (b) Figure 2 shows the chest clinic network. Are “visit to Asia” and “smoking” independent? Are they still independent once a positive X-ray is observed? Explain using the graph. (1 point)

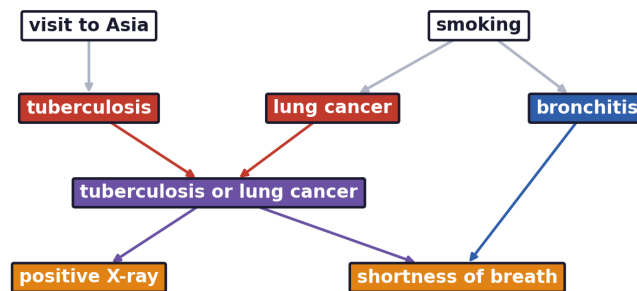


Figure 2: The chest clinic network. Every variable is binary.

- (c) Figure 3 shows a model of how the brain combines hearing and vision. What does the variable C represent, and why must the brain infer it before combining the sound and the sight? On trials that a subject reports as having two causes, the sound is heard pushed away from the light (a negative bias). Explain why. **(1 point)**

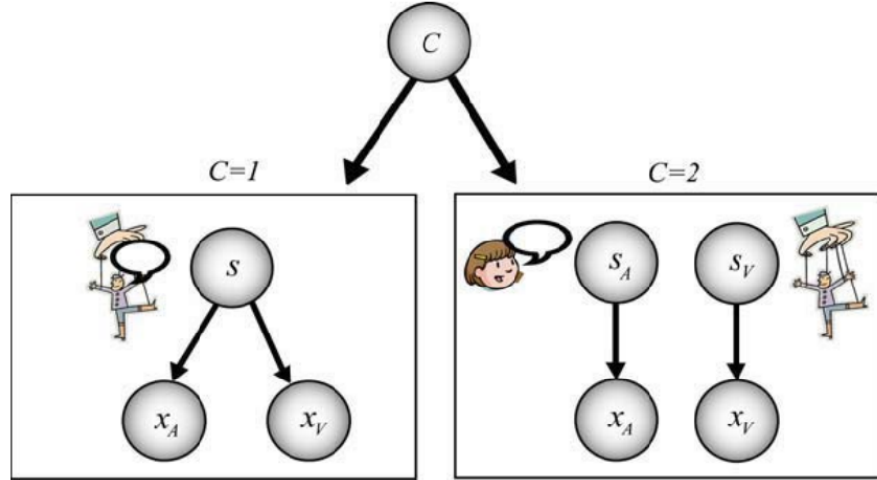


Figure 3: Hearing and vision. x_A and x_V are the auditory and the visual measurements of position; s , s_A and s_V are the positions of their sources.

2. **(4 points) Cue combination.** The slant S of a surface is estimated from texture (cue C_1) and from binocular disparity (cue C_2). The cues are conditionally independent given S , with Gaussian likelihoods $P(C_i | S) \propto \exp\{-(C_i - S)^2/2\sigma_i^2\}$, where $\sigma_1 = 6^\circ$ and $\sigma_2 = 3^\circ$. The measurements are $C_1 = 24^\circ$ and $C_2 = 36^\circ$.
- (a) With a uniform prior, compute the optimal estimate S^* , the weights of the two cues, and the standard deviation of the combined estimate. **(2 points)**
- (b) Now use the Gaussian prior $P(S) = \mathcal{N}(S; 0^\circ, 12^2)$, which favours surfaces facing the viewer. Compute the estimate from the joint posterior $P(S | C_1, C_2)$ and the two single-cue estimates $S_1^* = \arg \max_S P(S | C_1)$ and $S_2^* = \arg \max_S P(S | C_2)$. Average S_1^* and S_2^* with the weights from part (a) and compare the result with the joint estimate. Explain the difference. **(2 points)**

3. **(4 points) Explaining away.** A dark patch on the ground ($D = 1$) can be caused by a cast shadow ($H = 1$) or by dark paint ($M = 1$). The two causes are independent a priori, with $P(H = 1) = 0.2$ and $P(M = 1) = 0.1$, and

(H, M)	$(0, 0)$	$(1, 0)$	$(0, 1)$	$(1, 1)$
$P(D = 1 \mid H, M)$	0.05	0.9	0.9	0.95

A second observation, $O = 1$, says that an object which could cast the shadow is visible nearby, with $P(O = 1 \mid H = 1) = 0.8$ and $P(O = 1 \mid H = 0) = 0.1$. O depends on H only: given H , it is independent of D and M .

- (a) Draw the directed graph of the four variables. Which node is a common effect, and which is a common cause? **(1 point)**
 - (b) Compute $P(M = 1 \mid D = 1)$ and $P(H = 1 \mid D = 1)$. **(1 point)**
 - (c) Compute $P(M = 1 \mid D = 1, O = 1)$. Explain why observing O changes the belief about the paint, although O and M are not joined by an edge. **(2 points)**
4. **(2 points)** Write the joint distribution $P(H, M, D, O)$ of the network in part 3 as a product of one conditional distribution per node, and count its free parameters. How many would a general table over four binary variables need? Are H and M independent? Are they independent given D ?

Question 6: Markov random fields and mean field theory (12 points)

1. (4 points) Three binary pixels $x_1, x_2, x_3 \in \{0, 1\}$ form a chain with energy

$$E(\mathbf{x}) = \psi_1 x_1 + \psi_2 x_2 + \psi_3 x_3 + \psi_{12} x_1 x_2 + \psi_{23} x_2 x_3,$$

where $\psi_1 = -1.0$, $\psi_2 = 0.8$, $\psi_3 = -0.6$, $\psi_{12} = \psi_{23} = -1.5$, and $P(\mathbf{x}) = e^{-E(\mathbf{x})}/Z$.

- (a) Enumerate the eight configurations and compute Z , the MAP configuration and the three marginals $P(x_i = 1)$. The unary term of the middle pixel favours $x_2 = 0$: what is x_2 in the MAP configuration, and why? (2 points)
- (b) Run the mean field update $q_i \leftarrow \sigma(-\psi_i - \sum_{j \in N(i)} \psi_{ij} q_j)$, with $\sigma(u) = 1/(1 + e^{-u})$, updating q_1, q_2, q_3 in turn (each update using the latest values of the other two) and starting from $q_i = \sigma(-\psi_i)$, until it converges (a few sweeps). Compare the beliefs with the true marginals. Which are more confident, and why? (2 points)

2. (4 points)

- (a) Mean field theory approximates the Gibbs distribution $P(\mathbf{x}) = e^{-E(\mathbf{x})}/Z$ by a distribution $Q(\mathbf{x})$. What is $Q(\mathbf{x})$, and how is the Kullback–Leibler divergence used as a measure of similarity between P and Q ? (2 points)
- (b) For binary labels the mean field free energy is

$$F(\mathbf{q}) = \sum_i [q_i \log q_i + (1 - q_i) \log(1 - q_i)] + \sum_i \psi_i q_i + \sum_{(i,j)} \psi_{ij} q_i q_j,$$

where the last sum counts each edge (i, j) of the graph once. Derive the update of part 1(b) by setting $\partial F / \partial q_i = 0$. Why does the partition function Z never have to be computed? (2 points)

3. **(2 points) The weak membrane.** One edge of the weak membrane costs $\min(Ad^2, B)$, where $d = x_i - x_j$ is the difference of the two outputs. Take $A = 400$ and $B = 4$.
- (a) Above what value of $|d|$ is it cheaper to break the edge than to keep it? **(1 point)**
 - (b) The E-step of the EM algorithm sets the break probability $b_{ij} = 1/(1 + \exp\{-(Ad^2 - B)\})$. Compute b_{ij} for $d = 0.05$ and for $d = 0.3$, and explain how b_{ij} enters the M-step. **(1 point)**
4. **(2 points) DeepLab.** Why does DeepLab put a conditional random field (CRF) after the convolutional network? Why does a grid model with a constant Potts weight lose thin structures such as the branches of a tree, and how does the fully connected CRF with Gaussian edge weights avoid this?

Question 7: Learning with hidden variables

(6 points)

1. **(3 points) EM for a mixture of Gaussians.** Five datapoints $\{0, 1, 3, 5, 6\}$ are modelled by a mixture of two one-dimensional Gaussians with equal weights and known $\sigma = 1$. The current means are $m_1 = 1$ and $m_2 = 4$.
 - (a) E-step: compute the responsibilities $P(\omega_1 | x)$ of the first Gaussian for all five datapoints. **(1 point)**
 - (b) M-step: compute the new means. **(1 point)**
 - (c) What does one iteration of k-means do from the same means? How is k-means related to EM for this mixture? **(1 point)**
2. **(3 points) Boltzmann machines.** Binary neurons $S_i \in \{0, 1\}$ have the energy $E(\vec{S}) = -\frac{1}{2} \sum_{i,j} \omega_{ij} S_i S_j$ with symmetric weights $\omega_{ij} = \omega_{ji}$, and $P(\vec{S}) = e^{-E(\vec{S})/T} / Z$. The neurons are divided into observed neurons $\vec{S}_o$ and hidden neurons $\vec{S}_h$.
 - (a) What is the difference between the observed and the hidden neurons? What is the difference between a Boltzmann machine and a restricted Boltzmann machine? **(1 point)**
 - (b) Write the learning rule for the weights ω_{ij} in terms of the expected values of $S_i S_j$ under the clamped and the free (unclamped) distributions. Why is the clamped term called Hebbian and the free term anti-Hebbian? When does learning stop? **(1 point)**
 - (c) Why is it easier to learn the weights of a restricted Boltzmann machine? **(1 point)**

Questions 8–10: Experimental Section (38 points)

In this section you will run three Jupyter notebooks. They come with this PDF in the zip file of the homework, together with the data. Unzip the file and keep the folder structure, so that the folder `data`, which all three notebooks use, stays next to them. The Python packages are listed in `environment.yml`; with conda, run `conda env create -f environment.yml` and `conda activate homework-1`, then start `jupyter notebook` in that folder. Each notebook runs in less than a minute on a laptop.

We only expect to see the generated plots and your answers to the questions in the notebooks, included in your PDF. Each question in a notebook is followed by a code cell for your experiments and a text cell for your answers; when a question asks for several parameter values, write a loop or copy the relevant cell into that code cell, so that every requested figure appears in your PDF.

Question 8: Bayesian edge detection (14 points)

Lectures 3 and 4. Notebook: `bayesian_edge_detection.ipynb`.

8.1 the two class-conditional distributions (3); 8.2 decisions, risk and ROC curves (4); 8.3 several filters at one pixel (4); 8.4 domain transfer (3).

Question 9: Markov random fields and mean field theory (14 points)

Lectures 6–8. Notebook: `markov_random_fields.ipynb`.

9.1 the pairwise weight (3); 9.2 update schedules (3); 9.3 mean field and Gibbs sampling (3); 9.4 deterministic annealing (2); 9.5 the weak membrane (3).

Question 10: K-means, EM and Boltzmann machines (10 points)

Lecture 8. Notebook: `kmeans_em_boltzmann.ipynb`.

10.1 local minima of k-means (3); 10.2 EM for a mixture of Gaussians (3); 10.3 spherical k-means on image patches (2); 10.4 a Boltzmann machine that learns bars (2).